Many small carriers: the mean-field limit of competitive urban delivery under zone-time charges

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Abstract

Cities increasingly face many small delivery operators rather than a few large ones. We study the limit of a Cournot-congestion game among delivery carriers when the share of every carrier goes to zero, on a time-expanded zone graph in which carriers keep multi-stop tours and fleet budgets. The limit is a potential mean-field game whose equilibrium is a Wardrop equilibrium of the zone-hour market. We ask how fast the finite market approaches it, and what this means for a city that sets charges from observed zone totals. Four results answer the rate question. The aggregate gap is at most $\frac{1}{2}(\bar{s}\langle m, A\rangle)^{1/2}$, where $\bar{s}$ is the largest delivery one carrier can make at one zone-hour, and an explicit two-zone family attains this order, so no rate of order $\bar{s}$ with an instance-free constant exists once carriers' plans couple zones. When carriers' problems separate by zone-hour, the gap is at most one carrier's delivery, with constant one and equality for identical carriers; for identical carriers it is at most $\|A\|/N$ under any coupling; and for replicas of a fixed instance it is $O(1/n)$ with a constant governed by the conditioning of the carrier types. For the city, the observation-only charge converges to the marginal-cost charge at the rate of the aggregate gap of an auxiliary game, and its welfare loss is at most $\frac{1}{2}\bar{s}\langle m, Q^{\mathrm{opt}}\rangle$ for every instance. Cautious (Γ-maximin) carriers shift the limit like a per-delivery tax, which costs welfare of second order in the belief width. On nine cities with a stylised hourly profile, the gap falls like $N^{-0.88}$ (95% interval $[-0.90, -0.86]$) between 24 and 1536 carriers, about one largest carrier delivery throughout; the observation-only charge lowers welfare when there are few carriers and approaches the marginal-cost charge as their number grows. Algebraic cores of the theory are checked in Lean.

Keywords: mean-field games, Wardrop equilibrium, Cournot competition, congestion pricing, urban logistics, rate of convergence

1. Introduction

Parcel delivery in large cities is carried out by a mixture of a few large carriers and very many small operators: subcontracted van owners, regional couriers and gig drivers. A companion paper (Dang, 2026) (the *Omega paper* below) studied a city that charges a handful of large carriers playing a Cournot-congestion game in which each carrier solves a vehicle-routing problem. Its leader results (dual implementation pricing and aggregate-response pricing) are driven by market power: with few carriers, the corrective charge contains a subsidy for the deliveries that Cournot carriers withhold.

This paper asks what remains of that picture when the carriers are small. When every carrier's share goes to zero, each becomes a price taker and the market approaches a Wardrop equilibrium of the zone-time market, which is a potential mean-field game on a time-expanded graph (Lasry

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and Lions, 2007; Huang et al., 2006; Gomes et al., 2011; Guéant, 2015; Calderone and Sastry, 2017). Three questions follow. How fast does the finite market approach the limit, and with what constant? What happens to the city's observation-only charge? What happens to the parts of the Omega theory that depend on the number of carriers, namely the belief width that absorbs rounding errors and the vehicle-count regime in which pure equilibria may fail to exist?

Contributions. The paper's main contribution is a precise answer to the rate question for this class of games.

1. A general bound $\|A^{\mathrm{N}} - A^{\mathrm{W}}\|_m \le \frac{1}{2}(\sum_i \|a_i^{\mathrm{W}}\|_m^2)^{1/2} \le \frac{1}{2}(\bar{s}\langle m, A^{\mathrm{W}}\rangle)^{1/2}$, also for pure equilibria with lumpy plans (Theorem 4), and a two-zone family that attains it up to the factor $1/\sqrt{2}$ (Proposition 5). Consequently, for carriers whose tours or fleets couple zones, *no* bound of order $\bar{s}$ with an instance-free constant exists; the worst case is of exact order $\bar{s}^{1/2}$, the rate of Paccagnan et al. (2019).
2. Exact $O(\bar{s})$ bounds under structure: constant one for zone-separable carriers, for convexified and lumpy plans alike, with equality for identical carriers (Theorem 6); constant one in the M-norm for identical carriers under any coupling (Proposition 7); $O(1/n)$ for replicas with an instance-dependent constant (Theorem 8) that is explicit in the smooth case and degrades with the collinearity of the carrier types (Proposition 9).
3. For the city: the observation-only rule contracts at the instance-free rate of the Omega paper for both behaviours; its finite-market fixed point equals κ times the Nash aggregate of an auxiliary game, so it converges to the marginal-cost charge at the rate of an aggregate gap (Theorem 12); and its welfare loss is at most $\frac{1}{2}\bar{s}\langle m, Q^{\mathrm{opt}}\rangle$ for every instance (Theorem 13), so efficiency converges at rate $O(\bar{s})$ even where aggregates converge only at rate $O(\bar{s}^{1/2})$.
4. In the limit, Γ-maximin carriers act like a tax $\delta m \odot w$; the observation-only charge then overcorrects, at a welfare cost of at most $\frac{1}{2}\delta^2 \|m \odot w\|^2_{(m+\kappa)^{-1}}$ (Proposition 15); the belief width needed for credal absorption is $O(d\bar{s})$ (Corollary 16). Under vehicle-count congestion the limit has an equilibrium, and it is a weighted potential game exactly when dwell times are rank one (Proposition 18).
5. A computational study on the nine cities of the Omega paper, extended to a time-expanded graph with ten hourly periods and a stylised hourly profile, with up to 1536 carriers.

Scope. The limit fits gig couriers, subcontracted van operators and small regional carriers; it does not fit a market dominated by three to six large carriers, for which the Omega paper's atomic model, or a model with major and minor players (Huang, 2010), is the faithful one. We do not treat the major-minor model here. The hourly demand and speed profile is stylised: the repository has no hourly data, and we anchor the profile at published peak-hour congestion values and grade every assumption (Section 9).

2. Related work and what is new

Nash to Wardrop. Haurie and Marcotte (1985) showed that Nash–Cournot equilibria of routing games converge to Wardrop equilibria as players split into replicas, with a bound of the same square-root order as Paccagnan et al. (2019, Thm. 1(3)), who proved $\|\sigma(x^{\mathrm{N}}) - \sigma(x^{\mathrm{W}})\| \le (2RL_2/(\alpha M))^{1/2}$ for aggregative games with coupling constraints, and $O(1/M)$ bounds when the Nash or Wardrop operator is strongly monotone in the individual strategies (their Thm. 1(1)–(2)). Jacquot and Wan (2018) bound the squared distance on parallel networks linearly in the largest class mass, again a square-root rate for the distance. Cominetti et al. (2023) study unsplittable atomic games and their

Poisson limits. To our reading, no lower bound showing that the square-root order is attained had been given; Proposition 5 provides one, for linear profits where strong monotonicity in individual strategies fails. Theorem 6 (constant one, sharp) and Theorem 13(b) (uniform $O(\bar{s})$ welfare) are, as far as we found, not in print; both are short.

Mean-field games on networks and tolls. Existence and uniqueness of potential congestion MFGs on graphs are known (Guéant, 2015; Gomes et al., 2011); MDP congestion games and their Wardrop equilibria are due to Calderone and Sastry (2017); tolls that enforce constraints in MDP congestion games, including adaptive tolls with unknown costs on real ride-hail data, are in Li et al. (2019, 2023); marginal-cost tolls go back to Beckmann et al. (1956) and trial-and-error implementation to Yang et al. (2004). Cabannes et al. (2021) solve N-player dynamic routing games through their mean-field limit and observe accuracy beyond a few dozen vehicles, without a theorem on the gap or prices. Stackelberg MFGs are studied by Aurell et al. (2022) and pricing with a continuum of congested followers by Harks and Schedel (2024). Our leader results reuse these mechanisms; what is new is the exact identity for the finite-market fixed point and the instance-free welfare rate. Cournot MFGs (Chan and Sircar, 2015) and ride-hailing MFGs (Li et al., 2025) have no congestion network or tolls; robust MFG existence is known (Langner et al., 2024), and we use only the Γ-maximin surcharge. Solvers for potential MFGs (fictitious play, conditional gradient, mirror descent) exist (Cardaliaguet and Hadikhanloo, 2017; Lavigne and Pfeiffer, 2023; Pérolat et al., 2022); we reuse conditional gradient as the model of decentralised adjustment.

3. Model

3.1. The time-expanded zone graph

The city is divided into zones $z \in [Z]$ and the delivery day into periods $t \in [H]$ (in the experiments $H = 10$ hourly periods from 08:00 to 18:00). The *time-expanded zone graph* G_H has a node (z, t) for every zone and period, a travel arc $(z, t) \to (z', t')$ whenever a vehicle that leaves z in period t reaches z' in period $t' \geq t$ (travel times depend on the departure period), and a waiting arc $(z, t) \to (z, t + 1)$. Because every arc moves forward in time, G_H is acyclic.

A *tour* of a carrier is a path in G_H that starts and ends at the carrier's depot and makes deliveries at some of the nodes it visits. Delivering in zone z during period t adds to the deliveries q_{zt} of the carrier at that node. The coordinates of the interaction are the nodes $k = (z, t)$ touched by some tour; we write $\mathcal{K}$ for this set and $d = |\mathcal{K}| \leq ZH$. Vehicle-hours spent in zone z during period t are $r\, q_{zt}$, where r is the in-zone time per delivery (assumption C3 of Dang, 2026): congestion is caused by the time spent serving a zone, and inter-zone driving is priced through the tour's own cost only. This is the delivery-unit reduction of the Omega paper.

Demand and congestion. In coordinate k, the inverse demand for deliveries is $\alpha_k - b_k Q_k$ with $b_k > 0$, where Q_k is the total number of deliveries made at k by all carriers. A vehicle-hour at k costs $\gamma(\xi_k + c_k X_k)$ where $X_k = rQ_k$ is the congestion variable. Per delivery, a carrier therefore pays $\gamma r \xi_k$ (part of its private cost) plus $\kappa_k Q_k$ with $\kappa_k = \gamma c_k r^2 > 0$. Write

$$m_k = b_k + \kappa_k, \qquad M = \operatorname{diag}(m), \qquad \varrho = \max_k \kappa_k / m_k < 1.$$

For $v \in \mathbb{R}^d$ and a positive weight vector w we write $\|v\|_w^2 = \sum_k w_k v_k^2$; thus $\|v\|_m$ is the M-norm and $\|v\|_{m^{-1}}$ its dual norm.

3.2. Carriers, plans and their convexification

There are N carriers. Carrier i has a finite set $\mathcal{P}_i$ of *plans*: a plan is a multiset of tours that the carrier's fleet can operate in one day. Plan p has deliveries $q_i(p) \in \mathbb{R}^d_{\geq 0}$ and private profit $\pi_i(p) = \sum_k \alpha_k q_{ik}(p) - C_i(p)$, where $C_i(p)$ is its routing cost (driver time, fuel, fixed costs and the free-flow congestion term $\gamma r\langle \xi, q_i(p)\rangle$). A *small multi-stop carrier* is a carrier whose fleet is small compared with the market; the mean-field limit lets the share of every carrier go to zero while plans keep their routing structure.

For most results the plan set is convexified. A carrier with a fleet budget B_i (vehicle-hours) and a tour set $\mathcal{T}_i$ chooses fleet weights $y_i \in Y_i = \{y \in \mathbb{R}^{\mathcal{T}_i}_{\geq 0} : \langle w_i, y\rangle \leq B_i\}$, where w_{ij} is the duration of tour j; its deliveries are $a_i = L_i y_i$ with $L_i \geq 0$ entrywise (column j is the delivery vector of tour j per vehicle-hour) and its profit is $\pi_i(y_i) = \langle p_i, y_i\rangle$. The abstract assumptions under which the theory holds are:

(A1) Y_i is a nonempty compact convex subset of a Euclidean space, $\pi_i : Y_i \to \mathbb{R}$ is concave and upper semicontinuous, and $a_i = L_i y_i$ with L_i linear and $L_i y_i \in \mathbb{R}^d_{\geq 0}$ for $y_i \in Y_i$;

(A2) $m_k > 0$ for all k.

The fleet polytope with linear profits satisfies (A1). We write $A = \sum_i a_i$ (in delivery units $A = Q$) and

$$\bar{s} = \max_i \max_{y \in Y_i} \max_k (L_i y)_k$$

for the largest number of deliveries that one carrier can make at one coordinate; $\bar{s}$ is the *share* parameter of the limit.

Payoffs and equilibria. The city posts a charge $u \in \mathbb{R}^d$ per delivery. Carrier i's payoff is

$$J_i(y_i; y_{-i}) = \pi_i(y_i) - \langle a_i, M(A_{-i} + a_i)\rangle - \langle u, a_i\rangle, \qquad A_{-i} = \sum_{j \neq i} a_j. \tag{1}$$

A *Nash equilibrium* y^{N} is a profile in which each y_i maximises $J_i(\cdot\,; y_{-i})$: a carrier takes its own effect on the price and the congestion of the coordinates it serves into account (Cournot behaviour). A *Wardrop* (*mean-field*) *equilibrium* y^{W} is a profile in which each y_i maximises $\pi_i(y_i) - \langle a_i, MA + u\rangle$ with the aggregate $A = A(y^{\mathrm{W}})$ held fixed: carriers are price takers. The mean-field reading is the usual one: if carrier i is replaced by a continuum of identical infinitesimal agents whose plans sum to a point of Y_i, each agent's own effect vanishes and the equilibrium conditions of the continuum are those of y^{W}. On G_H this is the Wardrop equilibrium of a Markov-decision-process congestion game in which vehicles choose paths (Calderone and Sastry, 2017), with an elastic (Cournot) price added; it is also a discrete-time, finite-state potential mean-field game (Gomes et al., 2011; Guéant, 2015).

The welfare of a profile, with charges treated as transfers, is

$$\mathrm{SW}(y) = \sum_i \pi_i(y_i) + \sum_k \Big[-\tfrac{b_k}{2} Q_k^2 - \kappa_k Q_k^2 \Big] = \sum_i \pi_i(y_i) - \tfrac{1}{2}\|Q\|^2_{m+\kappa}, \tag{2}$$

the gross consumer benefit $\alpha_k Q_k - \frac{b_k}{2} Q_k^2$ (its linear part is inside π_i) minus routing and congestion costs. A *social optimum* maximises SW.

3.3. Potentials and a linearisation lemma

The first lemma is elementary convex analysis; it is used in almost every proof below.

Lemma 1 (Linearisation)**.** *Let C be convex, $h : C \to \mathbb{R}$ concave and g convex and differentiable on a neighbourhood of C. If x^* maximises $h - g$ over C, then x^* maximises $h - \langle \nabla g(x^*), \cdot \rangle$ over C.*

Proof. Let $x \in C$ and $x_t = x^* + t(x - x^*)$ for $t \in (0, 1]$. Optimality and concavity of h give $0 \geq h(x_t) - g(x_t) - h(x^*) + g(x^*) \geq t\,(h(x) - h(x^*)) - (g(x_t) - g(x^*))$. Divide by t and let $t \downarrow 0$: $0 \geq h(x) - h(x^*) - \langle \nabla g(x^*), x - x^* \rangle$. □

Lemma 2 (Potentials, uniqueness and the own surcharge)**.** *Assume (A1)–(A2) and fix u. Let*

$$\Phi_{\mathrm{N}}(y) = \sum_i \pi_i(y_i) - \langle u, A \rangle - \tfrac{1}{2}\|A\|_m^2 - \tfrac{1}{2}\sum_i \|a_i\|_m^2, \qquad \Phi_{\mathrm{W}}(y) = \sum_i \pi_i(y_i) - \langle u, A \rangle - \tfrac{1}{2}\|A\|_m^2.$$

(a) y is a Nash equilibrium iff it maximises Φ_{N} over $\prod_i Y_i$, and a Wardrop equilibrium iff it maximises Φ_{W}; a social optimum maximises Φ_{W} with $u = 0$ and m replaced by $m + \kappa$. All three sets are nonempty.

(b) The Nash contributions a_i^{N} are unique, and so is the Wardrop aggregate A^{W}; the Wardrop contributions need not be.

(c) (Own surcharge.) Every Nash equilibrium is a Wardrop equilibrium of the game in which carrier i pays the additional per-delivery charge $\sigma_i = Ma_i^{\mathrm{N}}$ (with a_i^{N} held fixed): each y_i^{N} maximises $\pi_i(y_i) - \langle a_i, MA^{\mathrm{N}} + u + Ma_i^{\mathrm{N}} \rangle$.

Proof. (a) $\Phi_{\mathrm{N}} - J_i$ does not depend on y_i: the terms of Φ_{N} containing y_i are $\pi_i - \langle u, a_i \rangle - \langle A_{-i}, Ma_i \rangle - \|a_i\|_m^2$, which are those of J_i. Hence a maximiser of Φ_{N} is a Nash equilibrium. Conversely let y be a Nash equilibrium and write $\Phi_{\mathrm{N}} = \sum_i \pi_i - f$ with f convex and differentiable. By Lemma 1 applied to carrier i (with $h = \pi_i$, $g = f(\cdot, y_{-i})$), $\pi_i(y_i') - \langle \nabla_i f(y), y_i' - y_i \rangle \leq \pi_i(y_i)$ for all $y_i' \in Y_i$. Summing over i and using $f(y') \geq f(y) + \langle \nabla f(y), y' - y \rangle$ gives $\Phi_{\mathrm{N}}(y') \leq \Phi_{\mathrm{N}}(y)$. The Wardrop case is the same with the linear functional $y_i \mapsto \langle a_i, MA + u \rangle$, whose y_i-gradient is $\nabla_i(\tfrac{1}{2}\|A\|_m^2 + \langle u, A \rangle)$. Existence: the potentials are upper semicontinuous on a compact set. (b) Φ_{N} is the sum of a concave function and $-\tfrac{1}{2}\|A\|_m^2 - \tfrac{1}{2}\sum_i \|a_i\|_m^2$, which is strictly concave in $(a_1, \ldots, a_N)$; two maximisers with different contributions would have a midpoint with a strictly larger value. The same argument with $-\tfrac{1}{2}\|A\|_m^2$ gives uniqueness of A^{W}. Non-uniqueness of Wardrop contributions: two identical carriers with linear profits can exchange deliveries. (c) Lemma 1 applied to J_i with $g(y_i) = \|L_i y_i\|_m^2$, whose gradient at y_i^{N} is $2L_i^\top Ma_i^{\mathrm{N}}$, and $A_{-i} + a_i^{\mathrm{N}} = A^{\mathrm{N}}$. □

Parts (a) and (b) are classical: the Nash potential of a Cournot game with linear demand is due to Slade (1994) and the Wardrop potential to Beckmann et al. (1956); for MDP congestion games see Calderone and Sastry (2017). Part (c) is the device behind all the bounds below: the finite game is the mean-field game in which each carrier pays a private surcharge equal to its own marginal effect Ma_i^{N}, a vector of size at most $\max_k m_k \cdot \bar{s}$.

Replicas. A *base type* θ is a triple $(\mathcal{Y}_\theta, \pi_\theta, L_\theta)$ satisfying (A1) with π_θ positively homogeneous (linear profits are). Its *n-split* consists of n carriers with $Y = \mathcal{Y}_\theta / n$, profit π_θ and the same L_θ: the fleet of the type is divided equally. A game whose carriers are the n_θ-splits of base types $\theta \in \Theta$ is a *replica game*.

Lemma 3 (Symmetry of replicas)**.** *In a replica game, carriers of the same type have the same Nash contributions, and the type totals $Y_\theta = \sum_{i \in \theta} y_i$ of a Nash equilibrium maximise*

$$\Phi^{(n)}(Y) = \sum_\theta \pi_\theta(Y_\theta) - \langle u, A \rangle - \tfrac{1}{2}\|A\|_m^2 - \tfrac{1}{2}\sum_\theta \frac{1}{n_\theta}\|L_\theta Y_\theta\|_m^2, \qquad A = \sum_\theta L_\theta Y_\theta,$$

over $\prod_\theta \mathcal{Y}_\theta$*. The Wardrop aggregate does not depend on* (n_θ)*.*

Proof. Φ_{N} is invariant under permutations of identical carriers, so by uniqueness of the Nash contributions (Lemma 2(b)) identical carriers have equal contributions. Averaging a maximiser over the permutations within each type gives a symmetric maximiser (concavity), whose value is $\Phi^{(n)}$ of its type totals; conversely every Y yields the symmetric profile $y_i = Y_\theta/n_\theta$. For the Wardrop part, Φ_{W} depends on y only through the type totals, and $\sum_{i\in\theta} \mathcal{Y}_\theta/n_\theta = \mathcal{Y}_\theta$ by convexity. □

Lemma 3 is also the computational reduction used in Section 8: the Nash equilibrium of n replicas per type costs one convex quadratic program in the type totals, whatever n is.

3.4. Imprecise beliefs and vehicle counts

Two extensions of the Omega model are kept. Under Γ-maximin beliefs of width δ about the rival aggregate (Omega, Lemma "Γ-maximin surcharge") each carrier evaluates its plans at the upper corner $A_{-i}+\delta w$ of a box of weights $w \in \mathbb{R}^d_{>0}$; Section 6 studies this layer in the limit. Under vehicle-count congestion (D2) the congestion variable counts vehicles instead of vehicle-hours; Section 7 treats it.

4. How far is the finite market from its mean-field limit?

Throughout this section the charge u is fixed (all bounds hold uniformly in u), y^{N} is a Nash equilibrium and y^{W} a Wardrop equilibrium of the same carriers, and $\Delta A = A^{\mathrm{N}} - A^{\mathrm{W}}$.

4.1. A general bound and its tightness

Theorem 4 (General bound)**.** *Assume (A1)–(A2).*

(a) For every Wardrop equilibrium y^{W}*,*

$$\|\Delta A\|_m^2 \le \sum_i \langle a_i^{\mathrm{N}}, M(a_i^{\mathrm{W}} - a_i^{\mathrm{N}})\rangle \le \frac{1}{4}\sum_i \|a_i^{\mathrm{W}}\|_m^2,$$

hence

$$\|\Delta A\|_m \le \frac{1}{2}\Big(\sum_i \|a_i^{\mathrm{W}}\|_m^2\Big)^{1/2} \le \frac{1}{2}\big(\bar{s}\,\langle m, A^{\mathrm{W}}\rangle\big)^{1/2}.$$

(b) (Lumpy plans.) Let p *be a pure Nash equilibrium of the game in which carrier* i *chooses a plan in the finite set* $\mathcal{P}_i$ *(payoff (1) with* $a_i = q_i(p_i)$*), and let* x^{W} *be a Wardrop equilibrium of its mixed extension (carrier* i *mixes over* $\mathcal{P}_i$*, contributes the mean* $\bar{a}_i$*). Then*

$$\|A(p) - A^{\mathrm{W}}\|_m^2 \le \sum_i \Big(\mathbb{E}_{x_i^{\mathrm{W}}}\|q_i\|_m^2 - \langle \bar{a}_i^{\mathrm{W}}, Mq_i(p_i)\rangle\Big) \le \bar{s}\,\langle m, A^{\mathrm{W}}\rangle,$$

with $\bar{s}$ *the largest delivery of a plan at a coordinate. A pure Nash equilibrium exists.*

Proof. (a) By Lemma 2(c), y_i^{N} maximises $\pi_i - \langle a_i, MA^{\mathrm{N}} + u + Ma_i^{\mathrm{N}}\rangle$, so it is at least as good as y_i^{W}; and y_i^{W} maximises $\pi_i - \langle a_i, MA^{\mathrm{W}} + u\rangle$, so it is at least as good as y_i^{N}:

$$\pi_i(y_i^{\mathrm{W}}) - \langle a_i^{\mathrm{W}}, MA^{\mathrm{N}} + u + Ma_i^{\mathrm{N}}\rangle \le \pi_i(y_i^{\mathrm{N}}) - \langle a_i^{\mathrm{N}}, MA^{\mathrm{N}} + u + Ma_i^{\mathrm{N}}\rangle,$$

$$\pi_i(y_i^{\mathrm{N}}) - \langle a_i^{\mathrm{N}}, MA^{\mathrm{W}} + u\rangle \le \pi_i(y_i^{\mathrm{W}}) - \langle a_i^{\mathrm{W}}, MA^{\mathrm{W}} + u\rangle.$$

Adding, the profits and charges cancel and $\langle a_i^{\mathrm{W}} - a_i^{\mathrm{N}}, M(A^{\mathrm{W}} - A^{\mathrm{N}})\rangle \le \langle a_i^{\mathrm{W}} - a_i^{\mathrm{N}}, Ma_i^{\mathrm{N}}\rangle$. Summing over i gives the first inequality. The second is $\langle x, M(y-x)\rangle \le \frac{1}{4}\|y\|_m^2$, i.e. $\|x - y/2\|_m^2 \ge 0$, with $x = a_i^{\mathrm{N}}$, $y = a_i^{\mathrm{W}}$. Finally $\sum_i \|a_i^{\mathrm{W}}\|_m^2 = \sum_k m_k \sum_i (a_{ik}^{\mathrm{W}})^2 \le \bar{s} \sum_k m_k A_k^{\mathrm{W}}$ because $0 \le a_{ik}^{\mathrm{W}} \le \bar{s}$.

(b) A pure equilibrium exists because Φ_{N} restricted to pure profiles is an exact potential of a finite game (Monderer and Shapley, 1996). At p, carrier i does not gain by deviating to any plan p'; averaging over $p' \sim x_i^{\mathrm{W}}$ (payoffs are evaluated against the realised rival aggregate $A_{-i} = A(p) - q_i(p_i)$),

$$\pi_i(p_i) - \langle q_i(p_i), M(A_{-i} + q_i(p_i))\rangle \ge \bar{\pi}_i(x_i^{\mathrm{W}}) - \langle \bar{a}_i^{\mathrm{W}}, MA_{-i}\rangle - \mathbb{E}_{x_i^{\mathrm{W}}} \|q_i\|_m^2.$$

The Wardrop condition, tested against the pure strategies p, gives $\sum_i \bar{\pi}_i(x_i^{\mathrm{W}}) - \langle A^{\mathrm{W}}, MA^{\mathrm{W}}\rangle \ge \sum_i \pi_i(p_i) - \langle A(p), MA^{\mathrm{W}}\rangle$ (the charge terms are omitted; they cancel as in (a)). Adding the sum over i of the first inequality to the second and expanding A_{-i} gives $0 \ge \|A(p) - A^{\mathrm{W}}\|_m^2 + \sum_i \langle \bar{a}_i^{\mathrm{W}}, Mq_i(p_i)\rangle - \sum_i \mathbb{E}_{x_i^{\mathrm{W}}} \|q_i\|_m^2$, which is the claim; the last bound uses $\langle \bar{a}_i^{\mathrm{W}}, Mq_i(p_i)\rangle \ge 0$ and $\mathbb{E}\|q_i\|_m^2 \le \bar{s}\,\langle m, \bar{a}_i^{\mathrm{W}}\rangle$. □

The first inequality of (a) is the summed variational-inequality argument of Paccagnan et al. (2019, Thm. 1(3)), written for this game; with $\bar{s} \sim 1/N$ it gives the rate $O(N^{-1/2})$ of that theorem. The constant $\frac{1}{2}$ and part (b) are additions. The next result shows that, without further structure, nothing better than $O(\bar{s}^{1/2})$ holds: the constant $\frac{1}{2}$ is attained up to the factor $1/\sqrt{2}$.

Proposition 5 (Tightness)**.** *Let $n \ge 2$ and $\varphi_n \in (0, \pi/2)$ with $1 - \cos\varphi_n = 1/n$; put $c = \cos\varphi_n$, $v = (c, \sin\varphi_n)$. Consider two coordinates with $M = I$ and $2n$ carriers: n carriers of type A choose $t \in [0, 1/n]$, deliver te_1 and earn $\alpha_A t$; n carriers of type B choose $t \in [0, 1/n]$, deliver tv and earn $\alpha_B t$; $(\alpha_A, \alpha_B) = (1, c)$. Then the Wardrop aggregate is $A^{\mathrm{W}} = e_1$ (type totals $T^{\mathrm{W}} = (1, 0)$), the Nash type totals are $T^{\mathrm{N}} = (\frac{3}{4} - \frac{1}{4n}, \frac{1}{4} - \frac{1}{4n})$, and*

$$\|A^{\mathrm{N}} - A^{\mathrm{W}}\|_2^2 = \frac{1}{8n} + \frac{2n-1}{8n^3}, \qquad \text{so} \qquad \|\Delta A\|_2 \ge \frac{1}{2\sqrt{2n}} = \frac{1}{\sqrt{2}} \cdot \frac{1}{2}\Big(\sum_i \|a_i^{\mathrm{W}}\|^2\Big)^{1/2}.$$

Every carrier can deliver at most $\bar{s} = 1/n$ at any coordinate, while $\|\Delta A\|_\infty \ge \|\Delta A\|_2/\sqrt{2} \ge \frac{1}{4}\sqrt{n}\,\bar{s}$. Hence there is no constant C, independent of the instance, with $\|A^{\mathrm{N}} - A^{\mathrm{W}}\| \le C\bar{s}$, even with two coordinates, linear profits and bounded data; over this class the worst-case gap is of exact order $\bar{s}^{1/2}$.

Proof. Write $J = [e_1, v]$ and $G = J^\top J = \left(\begin{smallmatrix} 1 & c \\ c & 1 \end{smallmatrix}\right) \succ 0$. By Lemma 3 it suffices to work with the type totals $T \in [0,1]^2$, $A = JT$. The Wardrop potential is $\langle \alpha, T\rangle - \frac{1}{2}T^\top GT$; its unconstrained maximiser $G^{-1}\alpha = (1, 0)$ lies in the box (on its boundary, where both partial derivatives $\alpha - GT$ vanish), so $T^{\mathrm{W}} = (1, 0)$ is the maximiser; it is unique because $G \succ 0$. The Nash potential of Lemma 3 is $\langle \alpha, T\rangle - \frac{1}{2}T^\top(G + \frac{1}{n}I)T$ (each type moves along a unit vector, so $\|L_\theta T_\theta\|^2 = T_\theta^2$); its unconstrained maximiser solves $(G + \frac{1}{n}I)T = \alpha$, and direct substitution (with $c = 1 - 1/n$) shows that T^{N} above solves it; it lies in the interior of the box, so it is the Nash equilibrium. Then $\Delta A = J(T^{\mathrm{N}} - T^{\mathrm{W}})$ with $T^{\mathrm{N}} - T^{\mathrm{W}} = -(\frac{1}{4} + \frac{1}{4n}, -\frac{1}{4} + \frac{1}{4n})$, and $\|\Delta A\|^2 = \Delta T^\top G \Delta T$ evaluates (using $\sin^2\varphi_n = 1 - c^2$) to the stated expression, which is at least $1/(8n)$. With the Wardrop selection in which every A carrier delivers $1/n$ and every B carrier nothing, $\sum_i \|a_i^{\mathrm{W}}\|^2 = 1/n$. Per-carrier deliveries are at most $1/n$ by construction. □

The mechanism is visible in the eigenvectors of G: when two types move along nearly parallel directions (φ small), their *difference* direction $e_1 - v$ is almost free for the mean-field market, so a small difference between the types' own surcharges (Ma_i^{N}, of size $1/n$) moves the type totals by $O(1/(n\lambda_{\min}(G)))$ in opposite directions, and the aggregate by $O(1/(n\lambda_{\min}(G)^{1/2}))$. For a *fixed*

angle φ the same computation gives $n\|\Delta A\|_2 \to 1/\sin\varphi$ as $n \to \infty$: the rate is $O(1/n)$, but with a constant that grows without bound as the types become collinear (Proposition 9 below states this in general). Proposition 5 answers the question of the memo negatively: an instance-free constant for the cross-zone case does not exist.

4.2. Exact $O(\bar{s})$ bounds under structure

Theorem 6 (Zone-separable carriers: constant one)**.** *Suppose every carrier's problem separates by coordinate: $Y_i = \prod_k Y_{ik}$, $\pi_i = \sum_k \pi_{ik}$, and a_{ik} depends on y_{ik} only, so that carrier i chooses a number of deliveries q_{ik} in a compact set $S_{ik} \subset \mathbb{R}_{\geq 0}$ (an interval for the convexified game, a finite set for lumpy plans) with concave (respectively arbitrary) profit π_{ik}. Then for every coordinate k and every charge u:*

(a) (convexified game) $0 \leq Q_k^{\mathrm{W}} - Q_k^{\mathrm{N}} \leq \max_i q_{ik}^{\mathrm{N}}$;
(b) (lumpy plans) every pure Nash equilibrium satisfies $0 \leq Q_k^{\mathrm{W}} - Q_k^{\mathrm{N}} \leq \max_i \max S_{ik}$, where Q^{W} is the Wardrop aggregate of the mixed extension.

In (a) equality holds when the N carriers are identical with linear profit and an interior equilibrium: then $Q_k^{\mathrm{N}} = \frac{N}{N+1} Q_k^{\mathrm{W}}$ and $Q_k^{\mathrm{W}} - Q_k^{\mathrm{N}} = Q_k^{\mathrm{W}}/(N+1) = q_{ik}^{\mathrm{N}}$.

Proof. Separability decouples the coordinates, so fix k, drop it from the notation and write $m = m_k$, $\pi_i = \pi_{ik}$, $Q = Q^{\mathrm{N}}$ and $q_i = q_i^{\mathrm{N}}$; the charge u_k is absorbed into π_i. Let $\hat{s}_i = q_i$ in case (a) and $\hat{s}_i = \max S_i$ in case (b). We first show two revealed-preference facts about the Nash choice q_i:

(i) for all $q' \in S_i$ with $q' < q_i$: $\pi_i(q_i) - mQq_i \geq \pi_i(q') - mQq'$;

(ii) for all $q' \in S_i$ with $q' > q_i$: $\pi_i(q_i) - m(Q + \hat{s}_i)q_i \geq \pi_i(q') - m(Q + \hat{s}_i)q'$.

In case (a), Lemma 2(c) says that q_i maximises $\pi_i(q) - m(Q+q_i)q$ over S_i, which implies (ii) directly and (i) because $m(Q+q_i)(q_i - q') \geq mQ(q_i - q')$ for $q' < q_i$. In case (b), the Nash condition against q' reads $\pi_i(q_i) - \pi_i(q') \geq m[(q_i - q')Q_{-i} + q_i^2 - q'^2] = m(q_i - q')(Q_{-i} + q_i + q')$ with $Q_{-i} = Q - q_i$; for $q' < q_i$ the right side is at least $m(q_i - q')Q$, and for $q' > q_i$ it is at least $-m(q' - q_i)(Q + \hat{s}_i)$ because $q' \leq \hat{s}_i$.

Now let $p \geq 0$ and let $\mathrm{BR}_i(p) = \arg\max_{q \in S_i} \pi_i(q) - pq$; in case (b) a Wardrop carrier mixes over $\mathrm{BR}_i(p)$, in case (a) it chooses a point of the interval $\mathrm{BR}_i(p)$; either way its delivery lies in $\operatorname{conv} \mathrm{BR}_i(p)$ and $Q^{\mathrm{W}} = \sum_i \bar{q}_i^{\mathrm{W}}$ with $\bar{q}_i^{\mathrm{W}} \in \operatorname{conv} \mathrm{BR}_i(mQ^{\mathrm{W}})$.

Upper bound $Q \leq Q^{\mathrm{W}}$. Suppose $Q > Q^{\mathrm{W}}$, so $p := mQ^{\mathrm{W}} < mQ$. For $q' < q_i$, (i) gives $\pi_i(q_i) - pq_i - (\pi_i(q') - pq') \geq (mQ - p)(q_i - q') > 0$, so no $q' < q_i$ is a best response at p, and $\bar{q}_i^{\mathrm{W}} \geq q_i$ for every i. Summing, $Q^{\mathrm{W}} \geq Q$, a contradiction.

Lower bound $Q \geq Q^{\mathrm{W}} - \hat{s}$, $\hat{s} = \max_i \hat{s}_i$. Suppose $Q < Q^{\mathrm{W}} - \hat{s}$, so $p := mQ^{\mathrm{W}} > m(Q + \hat{s}) \geq m(Q + \hat{s}_i)$. For $q' > q_i$, (ii) gives $\pi_i(q_i) - pq_i - (\pi_i(q') - pq') \geq (p - m(Q + \hat{s}_i))(q' - q_i) > 0$, so $\bar{q}_i^{\mathrm{W}} \leq q_i$ for all i and $Q^{\mathrm{W}} \leq Q$, a contradiction.

Equality case: with profit αq and an interior solution, the first-order conditions are $\alpha = mQ^{\mathrm{W}}$ and $\alpha = mQ^{\mathrm{N}} + mq$ with $Q^{\mathrm{N}} = Nq$, so $q = \alpha/(m(N+1))$ and $Q^{\mathrm{W}} - Q^{\mathrm{N}} = \alpha/m - N\alpha/(m(N+1)) = q$. □

The theorem needs no symmetry, no smoothness and no bound on the number of carriers, and the constant is attained. Its scope is a carrier whose plans in different zone-hours do not compete for the same fleet. A carrier whose fleet budget is shared across zones, or whose tours visit several zones, is not separable; Proposition 5 shows that for such carriers no analogue with an instance-free constant holds.

Proposition 7 (Equal own surcharges)**.** *If all carriers have the same Nash contribution $\bar{a}$ (in particular, if they are identical: Lemma 3), then $\|A^{\mathrm{N}} - A^{\mathrm{W}}\|_m \le \|\bar{a}\|_m = \|A^{\mathrm{N}}\|_m/N$, for any coupling across coordinates.*

Proof. In the first inequality of Theorem 4(a), $\sum_i \langle \bar{a}, M(a_i^{\mathrm{W}} - \bar{a})\rangle = \langle \bar{a}, M(A^{\mathrm{W}} - A^{\mathrm{N}})\rangle \le \|\bar{a}\|_m \|\Delta A\|_m$. □

Theorem 8 (Replicas: $O(1/n)$ with an instance-dependent constant)**.** *Consider a replica game with base types $\theta \in \Theta$, polyhedral $\mathcal{Y}_\theta$ and linear profits, and let $n = \min_\theta n_\theta$. There is a constant C, depending on the base instance (and on u) but not on (n_θ), such that $\|A^{\mathrm{N}} - A^{\mathrm{W}}\|_m \le C/n$ for all $n \ge 1$.*

Proof. For $\sigma = (\sigma_\theta)_\theta \in \mathbb{R}^{d\times\Theta}$ let $S(\sigma)$ be the Wardrop aggregate of the base types (one carrier per type) when type θ pays the extra per-delivery charge σ_θ; it is unique by Lemma 2(b), and $S(0) = A^{\mathrm{W}}$ for every replica game (Lemma 3). By Lemmas 3 and 2(c), the Nash aggregate of the replica game is $S(\sigma^n)$ with $\sigma^n_\theta = ML_\theta Y^{\mathrm{N}}_\theta/n_\theta$, so $\|\sigma^n\| \le c_1/n$ with $c_1 = \max_\theta \max_{Y\in\mathcal{Y}_\theta} \|ML_\theta Y\|$. The Wardrop equilibria under σ are the solutions of the Karush–Kuhn–Tucker system of a convex quadratic program with polyhedral constraints in which σ enters linearly; its solution set, as a multifunction of σ, has a graph that is a finite union of polyhedra. By Robinson (1981) such a multifunction is locally upper Lipschitz: there are $\eta, C_0 > 0$ with $S(\sigma) \in S(0) + C_0\|\sigma\|\mathbb{B}$ whenever $\|\sigma\| \le \eta$. Hence $\|\Delta A\| \le C_0 c_1/n$ for $n \ge c_1/\eta$, and $\|\Delta A\| \le D \le D(c_1/\eta)/n$ otherwise, where D is the diameter of the set of feasible aggregates. □

The constant of Theorem 8 is not explicit. In the smooth interior case it is:

Proposition 9 (Affine interior replicas: explicit constant)**.** *Let base type θ choose t_θ in an open set, deliver $L_\theta t_\theta$ and earn $\langle c_\theta, t_\theta\rangle - \frac{1}{2} t_\theta^\top K_\theta t_\theta$ with $K_\theta \succeq 0$ (its n_θ-split carriers earn $\langle c_\theta, t\rangle - \frac{n_\theta}{2} t^\top K_\theta t$). Let $L = [L_\theta]_\theta$, $G = L^\top ML$, $K = \mathrm{diag}(K_\theta)$, $D_n = \mathrm{diag}(L_\theta^\top ML_\theta/n_\theta)$, and suppose $G + K \succ 0$ and that both equilibria are interior. Then*

$$\begin{aligned}\|A^{\mathrm{N}} - A^{\mathrm{W}}\|_m &= \|G^{1/2}(G + K + D_n)^{-1} D_n T^{\mathrm{W}}\|_2 \le \|D_n T^{\mathrm{W}}\|_{(G+K)^{-1}} \\ &\le \lambda_{\min}(G + K)^{-1/2} \Big(\sum_\theta \|L_\theta^\top M a_\theta^{\mathrm{W}}\|^2 \Big)^{1/2},\end{aligned}$$

with $a_\theta^{\mathrm{W}} = L_\theta T_\theta^{\mathrm{W}}/n_\theta$ the per-carrier Wardrop delivery of type θ.

Proof. The first-order conditions are $(G + K)T^{\mathrm{W}} = c - L^\top u$ and $(G + K + D_n)T^{\mathrm{N}} = c - L^\top u$, so $\Delta T = -(G + K + D_n)^{-1} D_n T^{\mathrm{W}}$ and $\|\Delta A\|_m^2 = \Delta T^\top G \Delta T$. With $S = G + K + D_n \succeq G + K \succ 0$, $S^{-1} G S^{-1} \preceq S^{-1} \preceq (G + K)^{-1}$, which gives the first inequality; the second is $\|x\|_{(G+K)^{-1}} \le \lambda_{\min}(G + K)^{-1/2}\|x\|_2$ and $(D_n T^{\mathrm{W}})_\theta = L_\theta^\top M a_\theta^{\mathrm{W}}$. □

When the profit of each type is linear ($K = 0$), the constant is $\lambda_{\min}(G)^{-1/2}$: the gap is $O(1/n)$ exactly when the types' delivery directions are linearly independent in the M-geometry, and the constant degrades as they approach collinearity, as in Proposition 5 where $\lambda_{\min}(G) = 1 - \cos\varphi$. With strictly concave individual profits ($K \succ 0$) the constant is at most $\lambda_{\min}(K)^{-1/2}$; this is the regime of Paccagnan et al. (2019, Thm. 1(1)–(2)), whose $O(1/N)$ bounds use strong monotonicity of the Nash or Wardrop operator in the individual strategies. Routing plans with linear costs and fleet budgets do not provide such monotonicity, which is why Theorem 8 has to rely on polyhedrality.

Summary of the rate question. With $\bar{s}$ the largest share of one carrier at one coordinate: the gap is at most $\frac{1}{2}(\bar{s}\langle m, A^{\mathrm{W}}\rangle)^{1/2}$ always (Theorem 4), and this order is attained (Proposition 5); it is at most $\bar{s}$ per coordinate with constant one for zone-separable carriers (Theorem 6) and at most $\|A^{\mathrm{N}}\|_m/N$ for identical carriers (Proposition 7); for replicas of a fixed instance it is $O(1/n)$ with a constant that depends on the conditioning of the instance (Theorem 8, Proposition 9).

5. The city's charge in the limit

The city posts a per-delivery charge $u \in \mathbb{R}^d$ (by Omega Proposition "one instrument per zone" it can be realised as a uniform per-delivery charge plus non-negative congestion charges per vehicle-hour). Let $Q^{\mathrm{N}}(u)$ and $Q^{\mathrm{W}}(u)$ be the Nash and Wardrop aggregates under u (unique by Lemma 2). The *observation-only rule* of the Omega paper, in its form without the market-power subsidy, posts

$$u^{t+1} = (1-\theta)u^t + \theta\,\kappa \odot Q(u^t), \qquad \theta \in (0,1],$$

using only the observed aggregate: the corrective charge at the observed deliveries. In the mean-field limit the market-power subsidy of the Omega rule has no object, and this is the natural rule.

Lemma 10 (Responses are nonexpansive)**.** *For $X \in \{\mathrm{N},\mathrm{W}\}$ and charges u, u': $\|Q^X(u)-Q^X(u')\|_m^2 \le \langle u'-u, Q^X(u)-Q^X(u')\rangle$, hence $\|Q^X(u)-Q^X(u')\|_m \le \|u-u'\|_{m^{-1}}$.*

Proof. For $X = \mathrm{W}$ write the two optimality inequalities of Theorem 4(a) with the two equilibria $y = y^{\mathrm{W}}(u)$, $y' = y^{\mathrm{W}}(u')$: $\pi_i(y_i') - \langle a_i', MA+u\rangle \le \pi_i(y_i) - \langle a_i, MA+u\rangle$ and the symmetric one. Adding and summing over i gives $\|A-A'\|_m^2 + \langle A-A', u-u'\rangle \le 0$. For $X = \mathrm{N}$ the surcharges Ma_i, Ma_i' of Lemma 2(c) add $\sum_i \|a_i - a_i'\|_m^2 \ge 0$ to the left side. The second inequality follows by Cauchy–Schwarz in the dual pair $(\|\cdot\|_m, \|\cdot\|_{m^{-1}})$. □

Proposition 11 (Contraction and fixed point)**.** *Let $T_X(u) = \kappa \odot Q^X(u)$ for $X \in \{\mathrm{N},\mathrm{W}\}$.*

- *(a) $\|T_X(u) - T_X(u')\|_{\kappa^{-1}} \le \varrho\|u-u'\|_{\kappa^{-1}}$ with $\varrho = \max_k \kappa_k/m_k < 1$; the plain rule ($\theta = 1$) converges linearly with ratio ϱ.*
- *(b) The unique fixed point is $u_X^* = \kappa \odot \tilde{Q}_X$, where $\tilde{Q}_X$ is the aggregate of any maximiser of $\tilde{\Phi}_X$, defined as Φ_X with $u = 0$ and the coupling $\frac{1}{2}\|A\|_m^2$ replaced by $\frac{1}{2}\|A\|_{m+\kappa}^2$ (the own terms of Φ_{N} keep the weight m).*
- *(c) $\tilde{\Phi}_{\mathrm{W}} = \mathrm{SW}$, so $u_{\mathrm{W}}^* = \kappa \odot Q^{\mathrm{opt}}$ is the marginal-cost (Pigou) charge and the Wardrop equilibrium under u_{W}^* is socially optimal.*

Proof. (a) $\|\kappa \odot \Delta Q\|_{\kappa^{-1}}^2 = \sum_k \kappa_k \Delta Q_k^2 \le \varrho\|\Delta Q\|_m^2 \le \varrho\|\Delta u\|_{m^{-1}}^2 \le \varrho^2\|\Delta u\|_{\kappa^{-1}}^2$, using $\kappa_k \le \varrho m_k$ twice and Lemma 10. (b) Let $\tilde{y}$ maximise $\tilde{\Phi}_X$ and $\tilde{Q}$ its aggregate. Lemma 1, applied to the convex term $\frac{1}{2}\|A\|_\kappa^2$, shows that $\tilde{y}$ maximises $\tilde{\Phi}_X + \frac{1}{2}\|A\|_\kappa^2 - \langle \kappa \odot \tilde{Q}, A\rangle$, which is Φ_X with $u = \kappa \odot \tilde{Q}$. By Lemma 2(a), $\tilde{y}$ is an X-equilibrium under that charge, so $\kappa \odot \tilde{Q}$ is a fixed point; it is unique by (a). (c) $\frac{1}{2}\|A\|_{m+\kappa}^2 = \frac{1}{2}\|Q\|_{b+2\kappa}^2$ is the quadratic part of (2). □

Part (a) is the contraction theorem of the Omega paper ("the response is a proximal map"), restated for the time-expanded game and for both behaviours; part (c) is the classical marginal-cost toll (Beckmann et al., 1956), here reached by observation alone. Part (b) is what makes the finite-N analysis exact:

Theorem 12 (Charge convergence)**.** *Let $\tilde{Q}_{\mathrm{N}}, \tilde{Q}_{\mathrm{W}}$ be the Nash and Wardrop aggregates of the auxiliary game of Proposition 11(b) (coupling $m+\kappa$, own weight m). Then*

$$u_{\mathrm{N}}^* - u_{\mathrm{W}}^* = \kappa \odot (\tilde{Q}_{\mathrm{N}} - \tilde{Q}_{\mathrm{W}}), \qquad \|u_{\mathrm{N}}^* - u_{\mathrm{W}}^*\|_{\kappa^{-1}} = \|\tilde{Q}_{\mathrm{N}} - \tilde{Q}_{\mathrm{W}}\|_\kappa,$$

and consequently

(a) *in general,* $\|u^*_{\mathrm{N}}-u^*_{\mathrm{W}}\|_{\kappa^{-1}} \le \frac{1}{2}\tilde{\varrho}^{1/2}\big(\sum_i \|\tilde{a}^{\mathrm{W}}_i\|^2_m\big)^{1/2} \le \frac{1}{2}\big(\tilde{\varrho}\,\bar{s}\,\langle m, \tilde{Q}_{\mathrm{W}}\rangle\big)^{1/2}$ *with* $\tilde{\varrho} = \max_k \kappa_k/(m_k+\kappa_k)$*;*

(b) *for zone-separable carriers,* $0 \le u^*_{\mathrm{W},k} - u^*_{\mathrm{N},k} \le \kappa_k \max_i \tilde{q}^{\mathrm{N}}_{ik} \le \kappa_k \bar{s}$ *for every* k*;*

(c) *for replicas (polyhedral types, linear profits),* $\|u^*_{\mathrm{N}} - u^*_{\mathrm{W}}\| = O(1/n)$ *with an instance-dependent constant.*

Moreover, for any charge-to-charge perturbation: if T *is a* ϱ*-contraction with fixed point* u^* *and* u' *is a fixed point of a map* T' *with* $\|T'u' - Tu'\| \le \varepsilon$*, then* $\|u' - u^*\| \le \varepsilon/(1-\varrho)$*; with* $T = T_{\mathrm{W}}$*,* $T' = T_{\mathrm{N}}$ *this gives the memo's form* $\|u^*_{\mathrm{N}} - u^*_{\mathrm{W}}\|_{\kappa^{-1}} \le \|Q^{\mathrm{N}}(u^*_{\mathrm{N}}) - Q^{\mathrm{W}}(u^*_{\mathrm{N}})\|_\kappa/(1-\varrho)$.

Proof. The identity is Proposition 11(b). The auxiliary game is a game of the form of Section 3 with coupling $m' = m + \kappa$ except that each carrier's own term carries m instead of m'; Lemma 2(c) then gives own surcharges Ma_i instead of $M'a_i$, and the proofs of Theorems 4, 6 and 8 go through verbatim: in Theorem 4(a) one obtains $\|\Delta\tilde{Q}\|^2_{m'} \le \sum_i \langle \tilde{a}^{\mathrm{N}}_i, M(\tilde{a}^{\mathrm{W}}_i - \tilde{a}^{\mathrm{N}}_i)\rangle \le \frac{1}{4}\sum_i \|\tilde{a}^{\mathrm{W}}_i\|^2_m$, and $\|x\|^2_\kappa \le \tilde{\varrho}\|x\|^2_{m'}$ gives (a); in Theorem 6 the surcharge $m_k q_i \le m'_k q_i$ only strengthens (ii), so $0 \le \tilde{Q}_{\mathrm{W},k} - \tilde{Q}_{\mathrm{N},k} \le \max_i \tilde{q}^{\mathrm{N}}_{ik}$, which is (b) after multiplying by κ_k; (c) is Theorem 8. The perturbation statement: $\|u' - u^*\| = \|T'u' - Tu^*\| \le \|T'u' - Tu'\| + \|Tu' - Tu^*\| \le \varepsilon + \varrho\|u' - u^*\|$. □

So the finite-market fixed point lies *below* the Pigou charge in every coordinate when carriers are zone-separable, and the shortfall is at most one carrier's deliveries times the congestion slope. The rate of convergence of the charge is the rate of convergence of the aggregate in the auxiliary game; the factor $1/(1-\varrho)$ of the perturbation argument is not needed.

Theorem 13 (Welfare)**.** *Let* y^{opt} *be a social optimum with aggregate* Q^{opt} *and* $\mathrm{SW}^{\mathrm{opt}} = \mathrm{SW}(y^{\mathrm{opt}})$.

(a) *For every charge* v *and every Wardrop equilibrium* y *under* v *with aggregate* Q*,*

$$\mathrm{SW}^{\mathrm{opt}} - \mathrm{SW}(y) \le \langle v - \kappa \odot Q^{\mathrm{opt}}, Q^{\mathrm{opt}} - Q\rangle - \tfrac{1}{2}\|Q - Q^{\mathrm{opt}}\|^2_b \le \tfrac{1}{2}\|v - \kappa \odot Q^{\mathrm{opt}}\|^2_{b^{-1}}.$$

(b) *(Uniform* $O(\bar{s})$ *efficiency of the observation-only charge.) The Nash equilibrium under the finite-market fixed point* u^*_{N} *satisfies*

$$0 \le \mathrm{SW}^{\mathrm{opt}} - \mathrm{SW}(y^{\mathrm{N}}(u^*_{\mathrm{N}})) \le \tfrac{1}{2}\sum_i \|a^{\mathrm{opt}}_i\|^2_m - \tfrac{1}{2}\sum_i \|a^{\mathrm{N}}_i\|^2_m \le \tfrac{1}{2}\bar{s}\,\langle m, Q^{\mathrm{opt}}\rangle.$$

(c) *For every charge* u*,* $\big|\mathrm{SW}(y^{\mathrm{N}}(u)) - \mathrm{SW}(y^{\mathrm{W}}(u))\big| \le \frac{1}{2}\sum_i \|a^{\mathrm{W}}_i(u)\|^2_m + \|Q^{\mathrm{N}}(u) - Q^{\mathrm{W}}(u)\|_\kappa\, \|\bar{Q}(u) - \kappa^{-1} \odot u\|_\kappa$*, where* $\bar{Q}(u)$ *is the midpoint of the two aggregates.*

Proof. Write $P^v(y) = \sum_i \pi_i - \frac{1}{2}\|Q\|^2_m - \langle v, Q\rangle$, so $\mathrm{SW} = P^v + \langle v, Q\rangle - \frac{1}{2}\|Q\|^2_\kappa$. (a) y maximises P^v, which is the sum of a concave function and $-\frac{1}{2}\|Q\|^2_m$; by Lemma 1, y also maximises $\sum_i \pi_i - \langle v, Q\rangle - \langle MQ, \cdot\rangle$ evaluated at the aggregate, and the exact expansion of the quadratic term then gives $P^v(y) - P^v(y^{\mathrm{opt}}) \ge \frac{1}{2}\|Q - Q^{\mathrm{opt}}\|^2_m$. Hence $\mathrm{SW}^{\mathrm{opt}} - \mathrm{SW}(y) \le -\frac{1}{2}\|\Delta Q\|^2_m + \langle v, Q^{\mathrm{opt}} - Q\rangle - \frac{1}{2}\|Q^{\mathrm{opt}}\|^2_\kappa + \frac{1}{2}\|Q\|^2_\kappa$ with $\Delta Q = Q - Q^{\mathrm{opt}}$, and $\frac{1}{2}\|Q\|^2_\kappa - \frac{1}{2}\|Q^{\mathrm{opt}}\|^2_\kappa = \langle \kappa \odot Q^{\mathrm{opt}}, \Delta Q\rangle + \frac{1}{2}\|\Delta Q\|^2_\kappa$; with $m - \kappa = b$ this is the first inequality, and maximising over ΔQ gives the second. (b) By Proposition 11(b), $y^{\mathrm{N}}(u^*_{\mathrm{N}})$ maximises $\tilde{\Phi}_{\mathrm{N}} = \mathrm{SW} - \frac{1}{2}\sum_i \|a_i\|^2_m$; compare with y^{opt}. The last inequality uses $0 \le a^{\mathrm{opt}}_{ik} \le \bar{s}$. (c) $\mathrm{SW} = P^u - \frac{1}{2}\|Q - \kappa^{-1}\odot u\|^2_\kappa + \frac{1}{2}\|u\|^2_{\kappa^{-1}}$. Lemma 2 gives $0 \le P^u(y^{\mathrm{W}}) - P^u(y^{\mathrm{N}}) \le \frac{1}{2}\sum_i \|a^{\mathrm{W}}_i\|^2_m - \frac{1}{2}\sum_i \|a^{\mathrm{N}}_i\|^2_m$ (because y^{N} maximises $P^u - \frac{1}{2}\sum \|a_i\|^2_m$), and the difference of the two quadratic terms is $\langle Q^{\mathrm{N}} - Q^{\mathrm{W}}, \bar{Q} - \kappa^{-1} \odot u\rangle_\kappa$. □

Part (b) is uniform: unlike the aggregates (Proposition 5), the welfare of the observation-only charge converges at rate $O(\bar{s})$ with constant $\frac{1}{2}\langle m, Q^{\mathrm{opt}}\rangle$ for every instance. The reason is that welfare is second order in the aggregate error near the optimum. Part (b) does not say that the

charge helps when N is small: with few carriers the Cournot shortfall dominates and a charge that only adds the congestion cost can lower welfare (Section 10). For a fixed small N the Omega rule with its market-power subsidy is the appropriate one.

Corollary 14 (Price of anarchy and leader gain)**.** *Let* $\mathrm{PoA}_X = \mathrm{SW}^{\mathrm{opt}}/\mathrm{SW}(y^X(0))$ *(welfare positive) and* $G_X = \mathrm{SW}(y^X(u^*_X)) - \mathrm{SW}(y^X(0))$ *the gain of the observation-only charge. Then* $|\mathrm{PoA}_{\mathrm{N}} - \mathrm{PoA}_{\mathrm{W}}| \le \mathrm{PoA}_{\mathrm{N}}\mathrm{PoA}_{\mathrm{W}}\,|\mathrm{SW}(y^{\mathrm{N}}(0)) - \mathrm{SW}(y^{\mathrm{W}}(0))|/\mathrm{SW}^{\mathrm{opt}}$ *and* $|G_{\mathrm{N}} - G_{\mathrm{W}}| \le [\mathrm{SW}^{\mathrm{opt}} - \mathrm{SW}(y^{\mathrm{N}}(u^*_{\mathrm{N}}))] + |\mathrm{SW}(y^{\mathrm{N}}(0)) - \mathrm{SW}(y^{\mathrm{W}}(0))|$*; by Theorems 13 and 4–8 both tend to zero at rate* $O(\bar{s}) + O(\|Q^{\mathrm{N}}(0) - Q^{\mathrm{W}}(0)\|_\kappa)$*, that is* $O(\bar{s})$ *for separable carriers and replicas and* $O(\bar{s}^{1/2})$ *in the worst case.*

Proof. Direct from the definitions, $\mathrm{SW}(y^{\mathrm{W}}(u^*_{\mathrm{W}})) = \mathrm{SW}^{\mathrm{opt}}$ (Proposition 11(c)) and Theorem 13(b),(c) at $u = 0$. □

6. Cautious carriers in the limit

Following decision (4) of the memo, only two pieces of the imprecise-probability layer of the Omega paper are kept: the Γ-maximin surcharge and the scaling of the belief width needed for credal absorption. Fix weights $w \in \mathbb{R}^d_{>0}$ and a width $\delta \ge 0$; a carrier believes that the mean rival aggregate lies in the box $\{A' : |A'_k - A_{-i,k}| \le \delta w_k\}$.

Proposition 15 (Γ-maximin carriers)**.** *(a) (Omega, Lemma "Γ-maximin surcharge".) Because deliveries are nonnegative and the coupling is affine, the lower expectation of every plan is attained at the upper corner* $A_{-i} + \delta w$*. Hence the* Γ*-maximin equilibria, finite or mean-field, are the precise equilibria under the charge* $u + \delta\, m \odot w$*:* $Q^{X,\delta}(u) = Q^X(u + \delta m \odot w)$*.*

(b) $\|Q^{X,\delta}(u) - Q^X(u)\|_m \le \delta\|w\|_m$*; for zone-separable carriers,* $0 \le Q^X_k(u) - Q^{X,\delta}_k(u) \le \delta w_k$ *for* $X = \mathrm{W}$*.*

(c) The observation-only rule under Γ*-maximin carriers,* $u \mapsto \kappa \odot Q^{X,\delta}(u)$*, contracts with the same ratio* ϱ *(it needs no knowledge of* δ*); its fixed point* u^*_δ *satisfies* $\|u^*_\delta - u^*_X\|_{\kappa^{-1}} \le \varrho^{1/2}\delta\|w\|_m/(1-\varrho)$*.*

(d) In the mean-field game, the equilibrium under u^*_δ *loses at most* $\frac{1}{2}\delta^2\|m \odot w\|^2_{(m+\kappa)^{-1}}$ *of welfare. A city that knows* δw *and posts* $u^*_\delta - \delta m \odot w$ *recovers the optimum.*

Proof. (a) is the Omega lemma; in the time-expanded model the coordinates are zone-hours, and nothing else changes. (b) Lemma 10 with $u' = u + \delta m \odot w$: $\|\delta m \odot w\|_{m^{-1}} = \delta\|w\|_m$. In the separable case the proof of Theorem 6 with a common price increase $\delta m_k w_k$ in coordinate k (in place of the private surcharges) gives the coordinatewise statement: the Wardrop aggregate falls, and by at most the price increase divided by m_k. (c) The map is $T_X(\cdot + \delta m \odot w)$, a ϱ-contraction in $\|\cdot\|_{\kappa^{-1}}$ (Proposition 11). Its distance to T_X at any u is $\|Q^X(u + \delta m \odot w) - Q^X(u)\|_\kappa \le \varrho^{1/2}\|\cdot\|_m \le \varrho^{1/2}\delta\|w\|_m$; apply the perturbation statement of Theorem 12. (d) The carriers face $v = u^*_\delta + \delta m \odot w$ and $u^*_\delta = \kappa \odot Q$ at the fixed point, so in Theorem 13(a) $v - \kappa \odot Q^{\mathrm{opt}} = \kappa \odot \Delta Q + \delta m \odot w$ with $\Delta Q = Q - Q^{\mathrm{opt}}$, and the bound becomes $-\langle \delta m \odot w, \Delta Q\rangle - \frac{1}{2}\|\Delta Q\|^2_{b+2\kappa} \le \frac{1}{2}\delta^2\|m \odot w\|^2_{(b+2\kappa)^{-1}}$, with $b + 2\kappa = m + \kappa$. □

So cautious carriers behave like carriers facing a tax of $\delta m_k w_k$ per delivery; the observation-only city cannot tell this tax apart from cost, and over-charges by about that amount. The welfare cost is second order in δ and of Harberger form.

Corollary 16 (Belief width needed for absorption)**.** *Let* $\sigma_{\max}$ *be the largest range of one carrier's deliveries at one coordinate in units of* w*. The uniform credal-absorption bound of the Omega paper,* $\delta^E_0 \le \sigma_{\max} \min\{N - 1, d + 1, 1 + \sqrt{d \ln(2d)/2}\}$*, is at most* $(d+1)\bar{s}/\min_k w_k$*. For* n*-splits of a fixed instance* $\bar{s} \propto 1/n$*, so the width at which a rounded pure profile becomes an E-admissible equilibrium is* $O(d/n)$ *and vanishes in the limit; the* N*-free terms bind exactly when* $N > d + 1$*.*

Proof. $\sigma_{\max} \le \bar{s}/\min_k w_k$ and $\min\{\cdot\} \le d+1$. □

With $d = ZH$ coordinates the terms $d+1$ and $1+\sqrt{d\ln(2d)/2}$ become relevant only for very many carriers (for Cairo with 179 active zone-hours in the experiments, $N > 180$), which is the regime of gig couriers and small operators. The corollary is algebra on the Omega theorem; its content is the regime statement.

Proposition 17 (Time-of-day charges can move the worst case of travel times)**.** *In the Omega routing layer the cost of a route is nondecreasing in every arc travel time, and the upper prevision over a probability box with unknown dependence is attained at the upper (slowest) marginals. If instead a period-dependent charge is paid at the arrival time, monotonicity can fail: with one arc whose travel time T has a p-box containing all laws on $[0,2]$, and cost $T + U\cdot\mathbf{1}\{T<1\}$ with $U>2$, the upper prevision equals $\sup_{T\in[0,2]}(T+U\mathbf{1}\{T<1\}) = 1+U$ (approached as $T\uparrow 1$), whereas the slowest law ($T=2$) costs 2.*

Proof. The p-box contains every Dirac law on $[0,2]$, so the upper prevision is the supremum of the cost; on $[0,1)$ the cost is $T+U<1+U$ and tends to $1+U$, on $[1,2]$ it is at most $2<1+U$. □

When charges increase over the day (or are constant), the arrival-time cost is again nondecreasing in travel time and the Omega corner property is restored; with decreasing charges, the robust route cost must be computed over the whole box. In the experiments below travel times are deterministic, so this proposition limits the transfer of the Omega routing layer to the time-expanded model rather than affecting the computations.

7. Vehicle-count congestion

Under D2 the congestion variable counts vehicles. Let mean-field types $\theta \in [\Theta]$ choose exposures $\tau_\theta \in \mathcal{T}_\theta \subset \mathbb{R}^Z_{\ge 0}$ (vehicle-hours per zone; compact convex) with concave value V_θ, and let a type-θ vehicle dwell $h_{\theta z} > 0$ hours in zone z, so that it contributes $\tau_{\theta z}/h_{\theta z}$ vehicles. The congestion variable is $X_z = \sum_\theta \tau_{\theta z}/h_{\theta z}$ and a vehicle-hour in z costs $g_z X_z$, $g_z > 0$. A mean-field equilibrium is a profile in which each τ_θ maximises $V_\theta(\tau) - \langle \tau, g\odot X\rangle$ at the realised X. The coupling operator $F_\theta(\tau) = g\odot X(\tau)$ has Jacobian blocks $\partial F_\theta/\partial\tau_{\theta'} = \mathrm{diag}_z(g_z/h_{\theta' z})$.

Proposition 18 (Existence and the rank-one condition)**.** *(a) An equilibrium exists.*

(b) If $h_{\theta z} = \eta_\theta \zeta_z$ (rank one), the game is a weighted potential game with weights $1/\eta_\theta$ and concave potential $\Psi(\tau) = \sum_\theta \eta_\theta^{-1} V_\theta(\tau_\theta) - \frac{1}{2}\sum_z g_z\zeta_z X_z^2$; equilibria are the maximisers of Ψ, and X is unique when the V_θ are concave.

(c) Positive weights w_θ with $w_\theta\,\partial F_\theta/\partial\tau_{\theta'}$ symmetric for all θ,θ' exist iff $h_{\theta z}/h_{\theta' z}$ does not depend on z, i.e. iff h is rank one.

(d) With two types in one zone and $h_1 \ne h_2$, the symmetric part of the coupling matrix $g\left(\begin{smallmatrix} 1/h_1 & 1/h_2 \\ 1/h_1 & 1/h_2\end{smallmatrix}\right)$ has determinant $-g^2(1/h_1 - 1/h_2)^2/4 < 0$, so the unweighted operator is not monotone.

Proof. (a) Each type's best-response correspondence at a given X is nonempty, convex-valued and upper hemicontinuous (Berge), and X is continuous in τ; Kakutani's theorem on $\prod_\theta \mathcal{T}_\theta$ gives a fixed point. (b) $\partial\Psi/\partial\tau_{\theta z}$ of the coupling part is $-g_z\zeta_z X_z/(\eta_\theta\zeta_z) = -\eta_\theta^{-1} g_z X_z$, which is η_θ^{-1} times the marginal coupling cost of type θ; Ψ is concave and the equivalence of equilibria and maximisers follows as in Lemma 2(a); X is a linear image on which Ψ is strictly concave. (c) Symmetry means $w_\theta g_z/h_{\theta' z} = w_{\theta'} g_z/h_{\theta z}$, i.e. $h_{\theta z}/h_{\theta' z} = w_{\theta'}/w_\theta$ for all z. (d) Direct computation. □

In the finite game, deciding whether a pure equilibrium exists under D2 is strongly NP-complete (Omega, Theorem "NP-completeness under D2"); in the convexified mean-field game an equilibrium always exists, so the limit removes the existence problem. What the limit does not remove is non-monotonicity: with dwell times that are not rank one there is no weighting under which the game is a potential game.

Conjecture 19. *For rank$(h) \geq 2$ there are instances with several isolated mean-field equilibria, and computing one is PPAD-hard in the number of types.*

Neither part has been checked; no reduction has been designed. The rank-one condition is plausible for a fleet of one vehicle class (dwell time proportional to a zone factor) and implausible for mixed fleets of vans and cargo bikes.

8. Computation

Equilibria as quadratic programs. By Lemma 2, every object of the paper maximises a concave quadratic potential over a product of fleet polytopes: Nash (Φ_{N}), Wardrop (Φ_{W}), the social optimum (Φ_{W} with coupling $m + \kappa$) and the two observation-only fixed points (Proposition 11(b)). We introduce the deliveries a_i and the aggregate Q as variables, so that the Hessian is diagonal, and solve the program with the interior-point method Clarabel (Goulart and Chen, 2024) at relative gap 10^{-13}. The accuracy of each solution is measured by the Frank–Wolfe (duality) gap Γ of the potential, and $\|Q - Q^*\|_m \leq (2\Gamma)^{1/2}$ because the potential is 1-strongly concave in Q in the m-norm; every reported gap carries this error bound, which is at most 0.135 of the gap in the replica runs.

Replicas cost the same as one carrier per type. By Lemma 3 the Nash equilibrium of n replicas per segment is one program in the segment totals with own weight $1/n$; the experiments therefore reach $N = 1536$ at the cost of $N = 6$. Heterogeneous splits need one block per carrier.

Learning dynamics. The conditional-gradient (Frank–Wolfe) method with exact line search is the natural decentralised dynamics: in each round every carrier computes one best response to the current zone-hour prices (one oracle call: the tour with the largest profit per vehicle-hour) and the profile moves toward it. For potential MFGs this is fictitious play in its conditional-gradient form (Cardaliaguet and Hadikhanloo, 2017; Lavigne and Pfeiffer, 2023); its gap decreases at rate $O(1/k)$. It is included as the method a decentralised market would follow and as an independent check of the interior-point solutions (tests in `mfg/tests`).

Best responses on the time-expanded graph. A carrier's best response to prices is a longest-path problem on the acyclic graph G_H with a constraint that a zone is served at most once per tour. With at most three stops per tour, as here, the tours are enumerated (one ordering per stop set, the one with least travel time, for each start hour); with longer tours the best response is a prize-collecting elementary path problem, NP-hard in general, and the labelling algorithms of the Omega paper apply.

9. Data and stylised hourly profile

Cities. We use the nine processed cities of the Omega paper (zones, populations, zone-to-zone travel times on the major road graph, depot scores and the parameter table with grades A/B/C; Table 1). For each city and seed we draw six *segments* of small carriers. A segment has a depot (high depot score, at least 3 km apart), eight customer zones drawn around the depot by population times a gravity decay, lots per stop as in the Omega calibration, and a fleet budget in vehicle-hours equal to

Table 1: Replica study per city (amplitude 1, five seeds). d: active zone-hours; $\varrho = \max \kappa/m$; slope: log-log slope of $\|Q^{\mathrm{N}} - Q^{\mathrm{W}}\|_m$ in N for $N \geq 24$, mean over seeds with 95% t-interval; ratio: $\max_k |\Delta Q_k|/s_N$ with s_N the largest Nash delivery of one carrier at one zone-hour, median and maximum over N and seeds; PoA-1 in % of optimal welfare at $N = 6$ and in the limit; charge gap $\|u^*_{\mathrm{N}} - u^*_{\mathrm{W}}\|_{\kappa^{-1}}/\|u^*_{\mathrm{W}}\|_{\kappa^{-1}}$ at $N = 6$ and $N = 1536$.

City	d	ϱ	slope [95% CI]	ratio med	ratio max	PoA_6	PoA_∞	charge_6	charge_{1536}
New Delhi	290	0.209	-0.92 [-0.95, -0.90]	1.02	1.53	22.4	0.357	0.424	0.0036
Cairo	179	0.009	-0.77 [-0.81, -0.73]	0.97	1.65	9.6	0.001	0.266	0.0025
Moscow	312	0.074	-0.93 [-0.96, -0.90]	0.73	1.55	23.3	0.026	0.414	0.0023
Buenos Aires	321	0.019	-0.92 [-0.94, -0.89]	0.99	1.62	26.9	0.003	0.453	0.0039
New York	295	0.025	-0.78 [-0.87, -0.69]	0.72	1.86	18.4	0.004	0.337	0.0027
Mumbai	247	0.154	-0.90 [-0.92, -0.89]	1.14	1.62	19.8	0.224	0.403	0.0038
London	316	0.083	-0.92 [-0.95, -0.89]	0.98	1.23	23.1	0.201	0.440	0.0038
Paris	322	0.042	-0.86 [-0.90, -0.83]	1.06	1.68	17.7	0.037	0.363	0.0037
Berlin	346	0.033	-0.92 [-0.97, -0.87]	0.94	1.82	25.0	0.010	0.443	0.0033

the segment's daily parcels divided by drops per vehicle-day times the shift length. The segment's demand share is 1/6 of the zone's parcels. Tours start in any of the ten hours, visit one to three customer zones, deliver one lot per stop, and end at the depot by 18:00; this gives between 1476 and 3095 tours and between 176 and 359 active zone-hours per instance.

Calibration. As in the Omega paper: $\alpha_k = 1.5\times$ delivery fee, $b_k = \alpha_k/(2Q_k^{\mathrm{full}})$ with Q_k^{full} the slice's demand in the hour, in-zone hours per delivery $r = 0.6\times$ shift/drops, and $\kappa_k = \gamma c_k r^2$ with c_k the derivative of the BPR curve at the hourly congestion level, scaled by the slice multiplicity. A tour costs the hourly vehicle cost (wage plus fuel) times its duration plus one hour.

The hourly profile is an assumption. The congestion level $\mathrm{cong}(h)$ is the TomTom all-day value with two Gaussian rush-hour bumps (width one hour) centred at 08:30 and 17:30 whose heights reproduce the TomTom morning and evening peak values of each city (anchors graded A; the shape graded C; Moscow's peaks, absent from the index, imputed from the other cities' ratios, graded C). The demand share $\psi(h)$ has a late-afternoon bump of +60% and a late-morning bump of +30%, normalised to mean one (graded C; no data). An amplitude parameter scales all deviations from a flat profile (amplitude zero is the static model split into ten equal hours). Results that depend on the profile are reported as sensitivity analyses over the amplitudes 0, 0.5, 1 and 1.5, and nothing in the paper relies on the profile being realistic.

10. Computational study

Protocol. Main study: nine cities, five seeds, profile amplitude 1, replica splits $k \in \{1, 2, 4, \ldots, 256\}$ per segment ($N = 6k$ carriers, $N \leq 1536$), 405 instances. Heterogeneous splits (Dirichlet(2) fleet shares and individual routing-cost multipliers in $[0.85, 1.15]$): three seeds, $k \leq 16$, 135 instances, each compared with the Wardrop equilibrium of the same population. Sensitivity: amplitudes 0, 0.5, 1.5 (three seeds) and a congestion-heavy variant with $\kappa \times 10$ (three seeds, 243 instances). All runs are resumable scripts in `mfg/experiments`; results are in `mfg/results`. Confidence intervals are t-intervals over seeds of per-seed least-squares slopes.

E1: the gap behaves like one carrier's delivery. Figure 1 shows the relative gap $\|Q^{\mathrm{N}} - Q^{\mathrm{W}}\|_m/\|Q^{\mathrm{W}}\|_m$ against N. Over all cities and seeds the slope for $N \geq 24$ is -0.88 $[-0.90, -0.86]$, between -0.93 and -0.77 by city. The slope is not -1 because the share itself does not fall exactly like $1/N$: the largest delivery of one carrier, s_N, scales like $N^{-0.92}$ (segments deliver more in total as the market

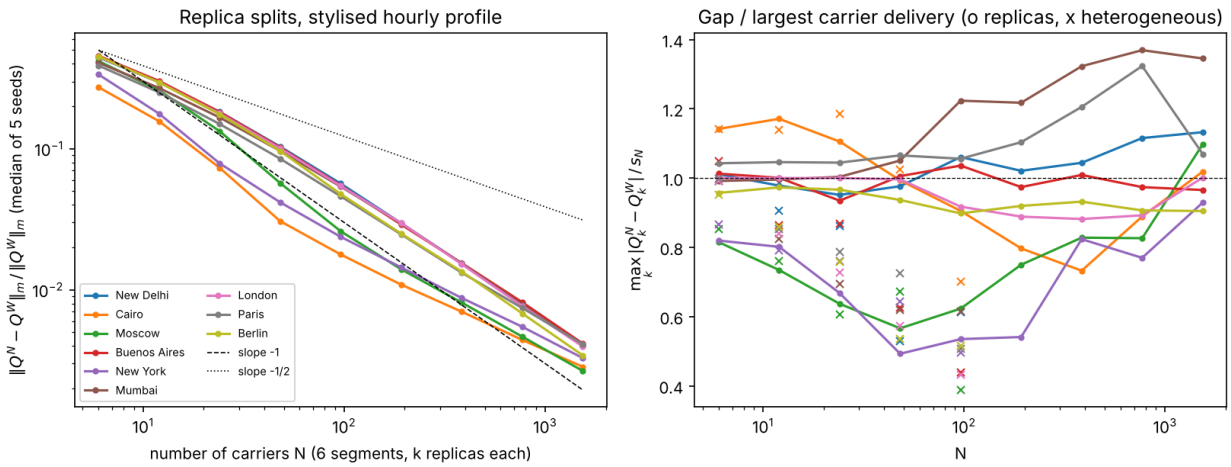


Figure 1: Left: relative aggregate gap against the number of carriers (replicas, median of five seeds). Right: largest coordinate gap in units of the largest carrier delivery s_N (circles: replicas; crosses: heterogeneous splits).

becomes more competitive), and the gap against s_N has slope 0.95 $[0.92, 0.97]$. In units of s_N the gap stays near one: median 0.98, range 0.35–1.86 (right panel). The general bound of Theorem 4 held in 405/405 instances (largest ratio 0.99). For heterogeneous splits the slope over $N = 12$ to 96 is -0.79 $[-0.83, -0.75]$, the ratio to s_N has median 0.76 and maximum 1.57, and the general bound held in 135/135. The heterogeneous slope is flatter and its range of N is short; we read it as consistent with $O(s_N)$ but not as evidence of a rate. No city comes close to the adversarial behaviour of Proposition 5: the theory says that such behaviour needs nearly collinear segment types with different own surcharges, and the city instances do not produce it.

E2: the charge and the welfare. The calibrated congestion is weak relative to demand: ϱ lies between 0.009 and 0.517, the mean-field price of anarchy is 0.026% of optimal welfare (median; maximum 0.591%), and the welfare cost of the finite market is almost entirely Cournot withholding: PoA-1 is 21.4% at $N = 6$ (range 8.7–28.1%) and 0.020% at $N = 1536$ (Figure 3). The observation-only fixed point approaches the Pigou charge with slope -0.91 $[-0.92, -0.89]$ (calibrated) and -0.90 $[-0.92, -0.89]$ (congestion $\times 10$, ϱ up to 0.914). The welfare bound of Theorem 13(b) held in 405/405 instances; the realised loss was at most 0.272 of the bound.

The negative result is the leader gain. With few carriers the observation-only charge *lowers* welfare: median -0.45% at $N = 6$ (calibrated) and -3.00% with congestion $\times 10$, because it taxes deliveries that Cournot carriers already withhold. With congestion $\times 10$ the gain is positive in every city and seed from $N = 192$ on and reaches 2.28% at $N = 1536$ against 2.36% in the limit. With the calibrated congestion the attainable gain is tiny (0.026% in the limit) and the charge never helps in all instances at once. The Omega flat rule, which adds the market-power subsidy, is the right rule for few carriers (E5).

E3: the hourly profile matters little for these questions. With amplitudes 0, 0.5, 1 and 1.5 the median PoA-1 at $N = 6$ is 20.7, 21.4, 21.6 and 21.8%, and the pooled gap slopes are -0.86, -0.87, -0.89 and -0.90 (Figure 4). The convergence results do not depend on the profile, as the theory predicts (the bounds are uniform in the data), and the economic quantities move by about one percentage point. This is a weak result in the sense that it says little about real time-of-day effects: with a stylised profile, it shows only that our conclusions are not artefacts of a particular shape.

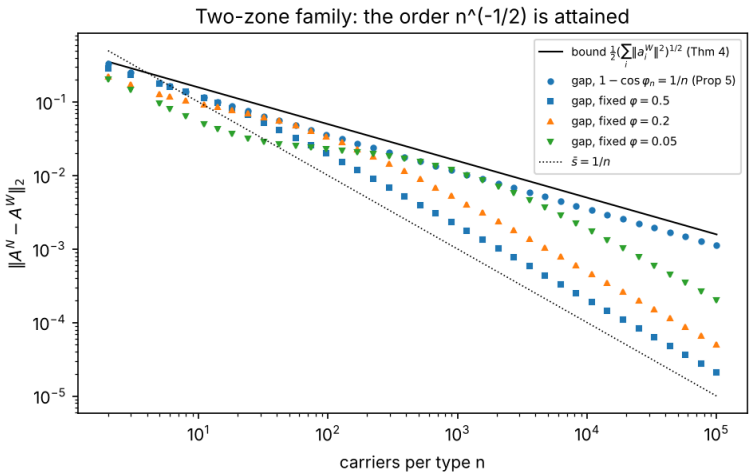


Figure 2: The two-zone family of Proposition 5. With $1-\cos\varphi_n = 1/n$ the gap follows the bound of Theorem 4 at ratio $1/\sqrt{2}$; for a fixed angle it eventually falls like $1/n$, with a constant $1/\sin\varphi$.

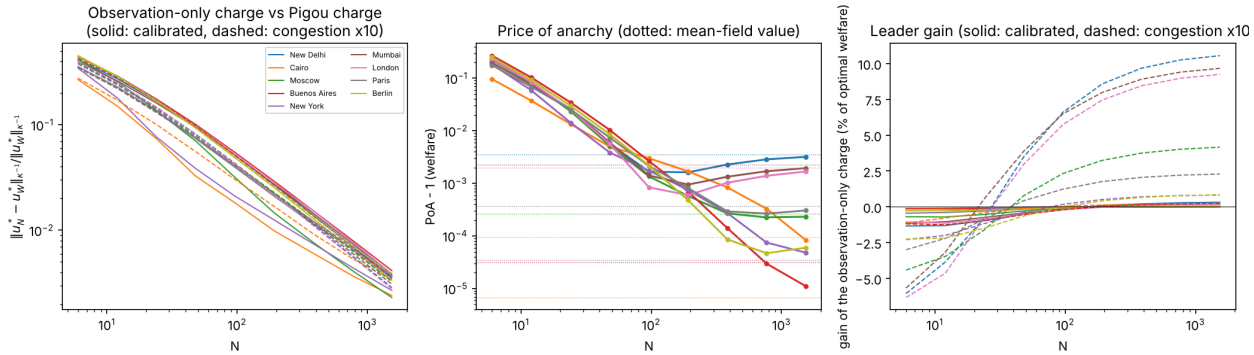


Figure 3: Left: distance of the finite-market observation-only charge from the Pigou charge. Middle: price of anarchy at N carriers (dotted: limit). Right: welfare gain of the observation-only charge (solid: calibrated congestion; dashed: congestion $\times 10$).

E4: cautious carriers and the belief width. With the belief box in units of the median lot, the mean-field aggregate moves by at most 0.973 of the bound $\delta\|w\|_m$ of Proposition 15(b) (median 0.614). The observation-only charge under cautious carriers loses 1.17% of optimal welfare at $\delta = 1$ and 11.85% at $\delta = 4$ (median), within the bound of Proposition 15(d) in 675/675 cases (median ratio 0.459); the charge shift satisfied part (c) in 675/675 cases (Figure 5). The absorption width of Corollary 16, relative to the mean demand of a zone-hour, falls from 481.21 at $N = 6$ to 0.698 at $N = 24576$ (Figure B.7 in the appendix); with d between 179 and 359 active zone-hours the N-free terms bind only beyond a few hundred carriers. This is a weak result: the uniform bound uses the largest delivery a carrier *could* make at one zone-hour (its whole fleet there), and even with twenty-four thousand carriers it remains of the order of a zone-hour's demand, so the absorption theorem does not give a practically small belief width on these instances.

E5: which observation-only rule. Figure 6 (right) compares the observation-only Pigou rule with the Omega flat rule $u \leftarrow \kappa Q - m \odot \bar{d}$, where $\bar{d}$ is the observed mean delivery per active carrier (damped iteration). At $N = 6$ the flat rule gains 16.41% (calibrated; 10.52% with congestion $\times 10$) where the Pigou rule loses -0.45% (-3.00%); at $N = 1536$ the flat rule gains 0.019% (2.284%) and

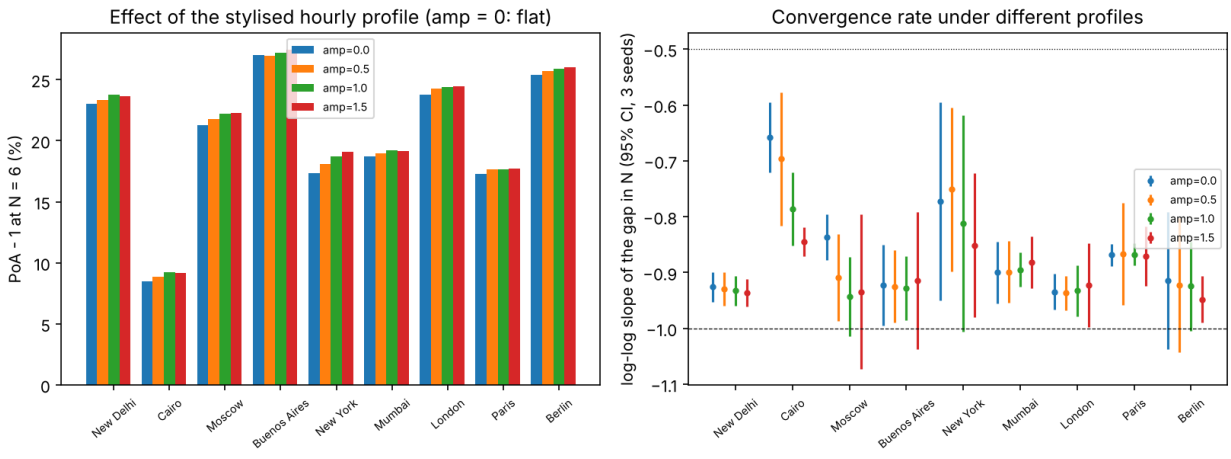


Figure 4: Sensitivity to the amplitude of the stylised hourly profile (three seeds). Left: PoA-1 at $N = 6$. Right: gap slopes with 95% intervals.

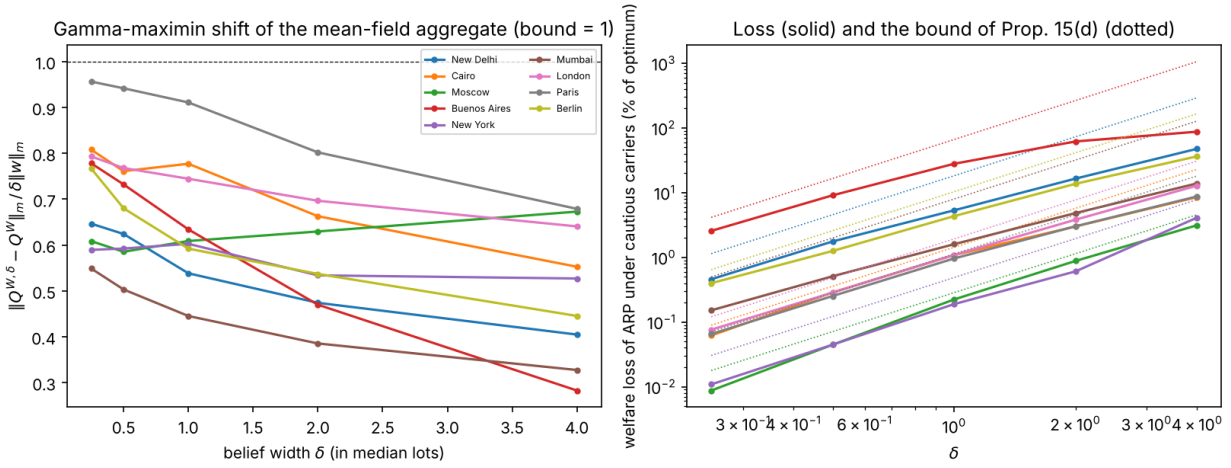


Figure 5: Γ-maximin carriers ($N = 1536$). Left: shift of the mean-field aggregate relative to the bound. Right: welfare loss of the observation-only charge and the bound $\frac{1}{2}\delta^2\|m \odot w\|^2_{(m+\kappa)^{-1}}$.

its charge is within 0.025 (relative, congestion $\times 10$) of the Pigou charge. The subsidy vanishes like the share, so the two rules meet in the limit. The plain rule contracted in every run at a per-round ratio of at most 0.945 times ϱ, counting the rounds before the distance reached 10^{-6} of its initial value, below which solver accuracy dominates (left panel).

E6: cost. The replica reduction makes the Nash equilibrium for $N = 1536$ as cheap as for $N = 6$ (well under a second per program). Heterogeneous splits grow with N: at $N = 96$ one interior-point solve took up to 90 seconds (Wardrop, whose solution set in the individual variables is not a singleton) and about 20 seconds (Nash). Plain Frank–Wolfe is slow: within 3000 rounds it reached a relative duality gap of 10^{-4} in 0/36 runs, and its relative gap after 3000 rounds was between 0.29% and 1.27% of the potential (Figure B.8 in the appendix). It is adequate as a model of how a decentralised market adjusts, not as a solver for the precision that the rate study needs; all reported equilibria come from the interior-point method.

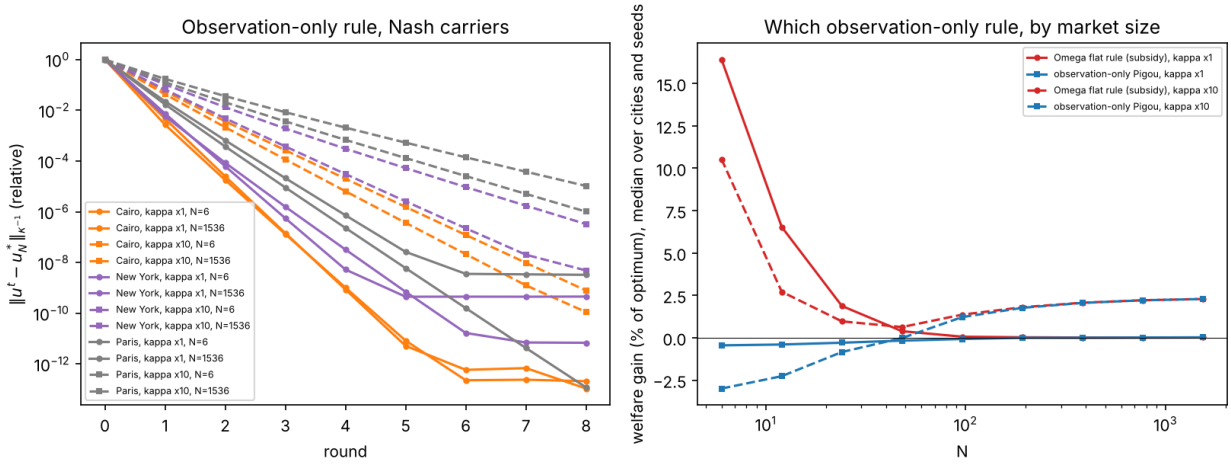


Figure 6: Left: plain observation-only iteration (Nash carriers), distance to the fixed point. Right: welfare gain of the Omega flat rule and of the observation-only Pigou rule against N (median over cities and seeds).

11. Discussion

What is proved, what is numerically supported, what is open. Proved, with complete proofs here: Lemmas 1–3, Theorems 4, 6, 8, 12, 13, Propositions 5, 7, 9, 11, 15, 17, 18 and Corollaries 14, 16. Theorem 8 rests on Robinson (1981), Proposition 15(a) and Corollary 16 on the Omega paper, and Proposition 11(a) restates the Omega contraction theorem. Every bound was also checked by brute force on small random games (Appendix Appendix A). Numerically supported only: that the city instances behave like $O(s_N)$ with a constant near one, and that heterogeneous splits do too. Open: an explicit constant for replicas in the polyhedral case, and Conjecture 19.

What the results mean for a city. With many small carriers, a city can set congestion charges from observed zone-hour totals and approach the marginal-cost charge; welfare converges at rate $O(\bar{s})$ regardless of how carriers' tours couple zones. With few carriers this rule is harmful, and the market-power correction of the Omega rule is needed. Cautious carriers make an observation-only city over-charge by about their caution margin; the welfare cost is second order.

Limitations. The hourly profile is stylised; real hourly demand and speed data would change the levels in E2–E4 and possibly the economic conclusions, but not the convergence statements. Travel times are deterministic in the experiments (the p-box routing layer of the Omega paper is not re-run on the time-expanded graph, and Proposition 17 shows that its corner property can fail with time-dependent charges). Congestion is linear in in-zone hours and ignores inter-zone driving. Tours have at most three stops. The segments are drawn, not observed. The mean-field limit removes market power, which is the economic content of the Omega leader results; for markets with a few large carriers and many small ones, a major-minor model is the right object and is left for later work.

12. Conclusion

The mean-field limit of competitive urban delivery is reached at rate $O(\bar{s})$ when carriers' problems separate across zone-hours, and in the city instances; in the worst case, once tours and fleets couple zones, only the square-root rate holds, and an explicit example attains it. The city's observation-only charge inherits these rates, but its efficiency converges at rate $O(\bar{s})$ for every

instance. Whether a constant that depends only on the number of zone-hours exists for natural subclasses (unweighted fleet budgets, for example) is the main open question.

Data and code availability

Code, instances, result files and the Lean development are in the repository (folder `mfg`).

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Appendix A. Numerical checks of the theory

The script `mfg/checks/theory_checks.py` computes, on small random games with linear profits and fleet budgets (and by exhaustive enumeration for pure equilibria of lumpy games), the slack of every inequality of Sections 3–7; its output is `theory_checks.out`. All slacks were nonnegative (up to the solver accuracy). In particular: the linearisation of Lemma 2(c) reproduces the Nash aggregate to 10^{-7}; the bound of Theorem 4(a) held in 150 games with largest ratio 0.92; the closed

form of Proposition 5 matches the solver; Theorem 6 held in 200 convexified and 300 lumpy one-zone games (358 pure equilibria), with the constant one attained; Theorem 4(b) held on 287 pure equilibria of multi-zone lumpy games.

Appendix B. Machine-checked cores

The Lean 4 file `mfg/lean/MFG/Core.lean` (Mathlib; no `sorry`; axioms `propext`, `Classical.choice`, `Quot.sound` only) checks: the symmetric closed form of Theorem 6; the inner-product algebra of Theorem 4(a) from the per-carrier optimality inequalities; the revealed-preference and sandwich steps of Theorem 6; the contraction perturbation lemma of Theorem 12; the weighted-norm chain of Proposition 11(a); the algebra of the tightness family (Proposition 5); the welfare inequalities behind Theorem 13(b) and Proposition 15(d); the determinant and the weight equivalence of Proposition 18; and the width bound of Corollary 16. It does not formalise existence, first-order optimality, Robinson's theorem or the Omega theorems that are used.

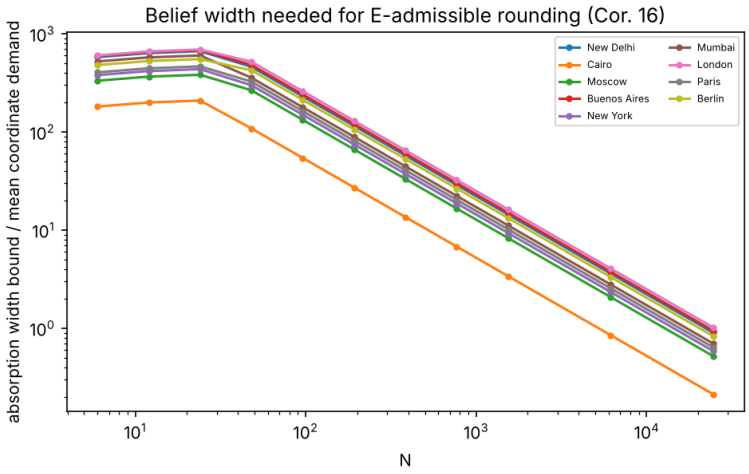


Figure B.7: Uniform belief width of Corollary 16 relative to the mean demand of a zone-hour (seed 0).

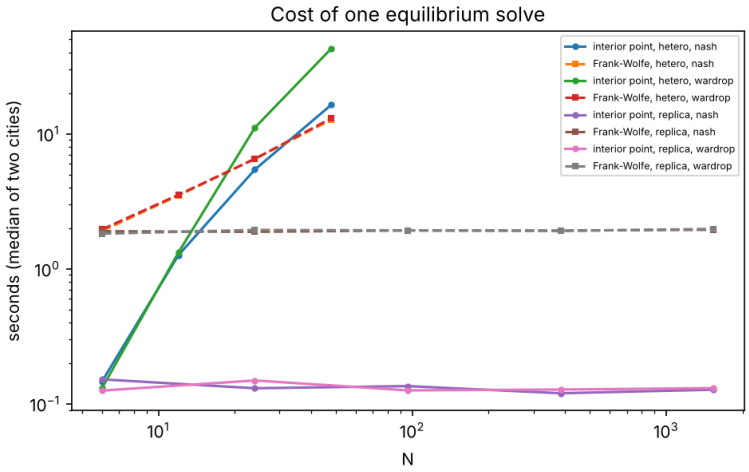


Figure B.8: Time of one equilibrium solve (interior point and Frank–Wolfe) for replicas and heterogeneous splits.